MIL-I-23322 (WEP)

29 October 1962

Superseding In Part
OS 9510
18 May 1959

MILITARY SPECIFICATION

IGNITER, MARK 173 MOD 1 (INERT ASSEMBLY

This specification has been approved by The Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope -This specification establishes the manufacturing requirements, acceptance and delivery of inert parts for Igniter, Mark 173 Mod 1.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications or such portions thereof as designated herein of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservations, Methods of
MIL-Q-9858	Quality Control System Requirements

FSC 1340

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

NAVY DEPARTMENT
BUREAU OF NAVAL WEAPONS

BuWeps LD 546907	Igniter Mark 173 Mod 1 (For Manufacture of Inert Parts)
BuWeps Dwg. 1990665	Ring Retainer
BuWeps Dwg. 1990666	Diaphragm, Safety
BuWeps Dwg. 1990667	Adapter, Igniter

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified documents and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications. -The following forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 with 1950 and 1957 Supplements	Screw Thread Standards for the Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite the latest edition and supplements thereto.)

UNIFORM CLASSIFICATION COMMITTEE Uniform Freight Classification Rules

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois. Orders for the above publication should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General Requirements. -The inert parts of Igniter Mark 173 Mod 1 shall meet all the requirements of this specification and the requirements specified in the drawings, specifications and other documents listed in LD 546907.

3.1.1 Materials. -Unless otherwise specified, all the materials and parts used in the Igniter shall conform strictly to the detailed requirements indicated on drawings, specifications and documents listed in LD 546907. When the drawings permit the use of equivalent material for proprietary items, authorization for a specific alternate shall be obtained from the procuring activity.

3.1.2 Finish. -All components shall be free from burrs and contamination which could result in malfunction of the equipment or be a safety hazard in handling.

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3.1.3 Dimensions. -Unless otherwise specified, all dimensions are to apply after all manufacturing processes, surface treatments, heat treating and testing have been completed. Dimensions apply prior to application of paint, decals, etc.

3.2 Safety Diaphragm Requirements.

3.2.1 Burst Pressure. -The diaphragm shall burst at the pressure specified on BuWeps drawing 1990666;

3.3 Preproduction Sample. -Unless otherwise specified in the contract or order, each manufacturer who has not produced the Igniter Mark 173 Mod 1 (Inert Assembly) previously shall submit a preproduction sample of 15 inert assembly igniters which have been manufactured using the methods and procedures proposed for the production lot. In addition, seven (7) safety diaphragms shall be submitted for the safety diaphragm burst test. This sample shall be tested as specified in Section 4 herein for the purpose of determining that the manufacturer is capable of producing a serviceable unit in conformance with the requirements of this section.

3.3.1 Renewal of Preproduction Sample. -In the event that production has been interrupted for more than six (6) months or if-there have been major changes which in the opinion of the Government Inspector or the procuring agency may adversely affect the characteristics of the Igniter, additional preproduction samples may be required.

3.4 Workmanship. -The workmanship exhibited in the preproduction sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become minimum production requirements for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission of the lot of inert igniters for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examintions and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any inspection deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot Size.

4.2.1 Safety Diaphragms. -A lot of safety diaphragms shall consists of all safety diaphragms having the same lot number.

4.2.2 Inert Assembly Igniters. -A lot of inert assembly igniters shall consist of all igniters having safety diaphragms with the same lot number.

4.3 Sampling. -All sample igniters or safety diaphragms shall be selected at random from assembled lots of igniters or safety diaphragms.

4.3.1 Sampling for General Requirement Inspection. -Sampling for general requirements inspection shall be in accordance with MIL-STD-105.

4.3.2 Sampling for Safety Diaphragm Burst Test. -Sampling for safety diaphragm burst tests shall be in accordance with MIL-STD-414, Table A-2 Inspection Level III. The minimum sample size is 5.

4.4 Quality Control. -Manufacturers quality control shall be in accordance with MIL-Q-9858-

4.4.1 Inspection and Tests.

4.401.1 Preproduction Sample Inspection and Tests. -The preproduction sample specified in 3.3 shall be forwarded to the activity designated by-the procuring activity for inspection and tests. All inert assembly igniters and safety diaphragms of the preproduction sample shall Be inspected for compliance with the general requirements of 3.1. All safety diaphragms of the preproduction sample shall be inspected and tested for compliance with the requirements of BuWeps drawing 1990666 and requirements of this specification. Ten (10) igniters will be loaded and test fired. Five (5) will be fired at the temperature of -40°F and five (5) at +140°F. Further production of the inert assembly igniters and safety diaphragms by the contractor prior to the completion of the pxeproduction sample inspection and tests or to approval by the procuring activity shall be at the manufacturer's risk. Acceptable inert assembly igniter preproduction sample units

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will be accounted as part of the quantity specified in the contract or order. If the inert parts manufacturer and the loading activity are the same, the inert parts preproduction sample may be tested simultaneously with the loaded igniter.

4.9.2 Production Inspection and Tests.

4.4.2.1 General Requirements Tests. -The contractor shall be responsible for meeting all requirements of 3.1. The applicable Ordance Classification of Defects (OCD) shall be required as the minimum acceptable inspection to be performed. OCD items of inspection shall be serialized and inspection data shall be traceable to a specific serial number.

4.4.2.2 Safety Diaphragm Burst Test. -Burst testing of sample safety diaphragms shall be with an apparatus conforming to BuWeps drawings 1990665 and 1990667 at the pressure area. Diaphragms shall be preformed prior to testing. Fluorescent penetrant inspection is not required.

4.5 Acceptance Criteria.

4.5.1 Preproduction Sample Acceptance. -Failure of any inert assembly igniter or safety diaphragm to meet any requirement may result in rejection of the sample and disqualification of the contractor. Final acceptance of the preproduction sample shall be made by the Chief of the Bureau of Naval Weapons.

4.5.2 Production Acceptance.

4.5.2.1 General Requirements. -The acceptable quality level (AQL) for general requirements inspection, using OCD's as minimum acceptable inspection to be performed are recorded as follows:

1. Plan A for lots 1 to 25
   (a) Critical 100% inspection
   (b) Major A 100% inspection
   (c) Major B AQL .25 percent
   (d) Minor A AQL 1.00 percent

2. Plan B for lots over 25
   (a) Critical 100% inspection
   (b) Major A AQL .25 percent
   (c) Major B AQL 1.00 percent
   (d) Minor A AQL 2.50 percent

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4.5.2.2 Safety Diaphragm Burst Test. -Based on MIL-STD-414, acceptable quality level (AQL) for the safety diaphragm burst test is .25 percent. If a minimum sample size of five is used, an M value of .422 shall apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. -Not applicable.

5.1.2 Level B. -Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservation application. -Not applicable.

5.1.3.2 Unit packaging. -Unit packaging of the igniters shall be in accordance with Method III of Specification MIL-P-116.

5.1.3.3 Intermediate Package. -Not applicable.

5.2 Packing.

5.2.1 Level A. -Not applicable.

5.2.2 Level B. -Not applicable.

5.2.3 Level C. -Eight (8) inert assembly igniters packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings.

5.3.2 Normal markings. -In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended Use. -The Igniter, Mark 173 Mod 1 Inert Assembly, when loaded, is intended for use in the Mark 23 Mod 0 Rocket Motor.

6.2 Value Engineering. -All manufacturers of items covered by this specification are encouraged to submit value engineering proposals to the procuring activity in an effort to reduce cost.

6.3 Ordering Data. -Requests, contracts, or orders should specify the following:

(a) Title and number of this specification
(b) Preproduction sample if different than specified in 3.3
(c) Sampling if different than specified in 4.3
(d) All applicable drawings and the latest revisions
(e) Packaging if different than specified in Section 5

6.3.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116.

6.4 Conflicting Requirements. -Conflicting requirements arising between this specification or any specification, publications, or drawings listed herein, shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

6.5 Security. -All parts or assemblies which are classified shall be stored and transported in accordance with appropriate security instructions.

606 Requests for Deviation. -Requests for deviation from this specification, applicable drawings, specifications, materials, or processes shall be forwarded to the procuring activity prior to manufacturer of the affected item or items. All deviations shall be Limited to the contract under which they were granted.

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6.7 Request for Waiver. -Requests for waiver from this specification, applicable drawings, specifications, materials, or processes shall be forwarded to the procuring activity. A waiver request shall include the following:

a. Exact nature of defect
b. OCD number and defect classification
c. Quantity of items involved

Notice. -When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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